THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 8: **Playing with Constructions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In rectangle ABCD, AC = BD = 18 cm. If O is the intersection, the sum OA + OB + OC + OD (cm) is: 1

(a) 36 **(b)** 45 **(c)** 54 **(d)** 27

2. Three circles of radius 7 cm are drawn in a triangle arrangement, each touching the other two. The triangle joining the centres has perimeter (cm): 1

(a) 63 **(b)** 49 **(c)** 42 **(d)** 35

3. An isosceles triangle OAB with OA = OB has $\angle AOB = 90^\circ$. Each base angle (degrees) is: 1

(a) 45 **(b)** 55 **(c)** 50 **(d)** 40

4. The diameter of the circle that exactly fits inside a square of side 18 cm is (cm): 1

(a) 18 **(b)** 17 **(c)** 19 **(d)** 27

5. Two concentric circles have radii 3 cm and 8 cm. A point is 0 cm from the common centre. It lies in the ring between the circles only if its distance is more than 3 cm and less than 8 cm. How many of the distances 2, 3, 5, 8, 7, 9 cm give a point in the ring? 1

(a) 2 **(b)** 6 **(c)** 3 **(d)** 4

6. A rectangle's diagonal is 26 cm. The perimeter of triangle AOB where OA = OB = 13 cm and AB = 10 cm is (cm): 1

(a) 36 **(b)** 35 **(c)** 49 **(d)** 37

7. A circle of radius 10 cm is drawn. A square of side equal to the diameter is drawn around it. The area of the square (cm^2) is: 1

(a) 420 **(b)** 410 **(c)** 390 **(d)** 400

8. Two circles of equal radius 9 cm touch externally. The distance from the outer end of one to the outer end of the other along the line of centres (cm): 1

(a) 54 **(b)** 27 **(c)** 45 **(d)** 36

9. **Assertion (A):** In rectangle ABCD the diagonals meet at O and AC = 10 cm. Then OA = 5 cm. 1
Reason (R): The diagonals of a rectangle are equal and bisect each other.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The perpendicular bisector of a segment AB cuts AB at its midpoint at right angles. 1
Reason (R): Every point on the perpendicular bisector of AB is at the same distance from A and from B.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A square ABCD of side 18 cm has diagonals meeting at O. Find the area of triangle AOB and the sum of areas of the four triangles. 2

12. A circle has centre O and radius 9 cm. Points are at distances 4 cm, 9 cm, 14 cm and 10 cm from O. Find how many lie inside, on and outside the circle. 2

13. A rectangle has diagonal intersection O with $\angle AOB = 100^\circ$. Find all four angles at O. 2

14. A square ABCD of side 22 cm has diagonals meeting at O. Find the area of triangle AOB and the sum of areas of the four triangles. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Four circles of radius 7 cm are placed at the corners of a square so that adjacent circles touch. Find the side of the square joining centres and its area. 3

16. Two concentric circles have radii 8 cm and 13 cm. Find the width of the ring, the diameter of the inner circle and the length of the longest chord of the outer circle. 3

17. In rectangle ABCD, AB = 20 cm, BC = 14 cm. The diagonals meet at O. If M is the midpoint of AB and N is the midpoint of BC, find BM and BN, and the perimeter and area of the rectangle MBNO. 3

OR

Four circles of radius 8 cm are placed at the corners of a square so that adjacent circles touch. Find the side of the square joining centres and its area. 3

18. Two concentric circles have radii 3 cm and 14 cm. Find the width of the ring, the diameter of the inner circle and the length of the longest chord of the outer circle. 3

OR

In rectangle ABCD, AB = 12 cm, BC = 8 cm. The diagonals meet at O. If M is the midpoint of AB and N is the midpoint of BC, find BM and BN, and the perimeter and area of the rectangle MBNO. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Two villages A and B are shown 8 cm apart on a map. The school must be built at a place that is the same distance from both villages. Ria draws the perpendicular bisector of AB with a compass and a ruler. 5

(i) Which construction does Ria use?

(ii) Where does the bisector cut AB and how far is that point from A?

(iii) What angle does the bisector make with AB?

(iv) Why must the compass opening be more than 4 cm? If the school P on the bisector is 5 cm from A, how far is it from B?

20. Describe the construction of a rectangle with diagonal 30 cm and one side 18 cm (you may use a compass for arcs). Find the other side if it is 24 cm, check using the property that the half-diagonal is 15 cm, and find the perimeter. 5

OR

A designer draws 10 identical circles of radius 4 cm in a row, each touching the next. Find the total length of the row and the distance between the centres of the first and the last circle. How many circles of this size fit exactly in a strip 96 cm long? 5

END OF QUESTION PAPER — Total: 40 Marks